Theory of Automata
CS-300
Lecture 10

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Objectives

- Definition of non-Regular Languages
- Theorem 13
  - Proof & Example
- Theorem 14
  - Proof & Example

Introduction

- By using FAs and regular expressions, we have been able to define many languages. Although these languages have many different structures, they take only a few basic forms:
  - Languages with required substrings
  - Languages that forbid some substrings
  - Languages that begin or end with certain substrings
  - Languages with certain even (or odd) properties, and so on.

Introduction

- We now turn our attention to some new forms, such as the language PALINDROME, or the language PRIME of all words a^p, where p is a prime number.

- We shall see that neither of these is a regular language. We can describe them in English, but they can not be defined by an FA. We need to build more powerful machines to define them.

Definition of Non-Regular Language

Definition:

- **A language that cannot be defined by a regular expression is called a non-regular language.**

Notes:

– By Kleene's theorem, a non-regular language can also **not** be accepted by any FA or TG.

– All languages are either regular or non-regular; none are both.

Case Study

- Consider the language

  L = {Λ; ab; aabb; aaabbb; aaaabbbb; aaaaabbbbb; …}

- The language L can also be written as

  L = {$a^n b^n$ for n = 0; 1; 2; 3; …}

  or for short

  L = {$a^n b^n$}

- Note that although L is a subset of many regular languages, such as a*b*; the language defined by ab also includes such strings as aab and bb that are not in L.

Example 1:

- We shall show that the language L = $\{a^n b^n\}$ is non-regular.
- Suppose on the contrary that L were regular. Then there must exist some FA that accepts L.

- Just for the sake of argument, let us assume that this FA has 95 states.

- We know that this FA must accept the word $a^{96}b^{96}$.

Example 1:

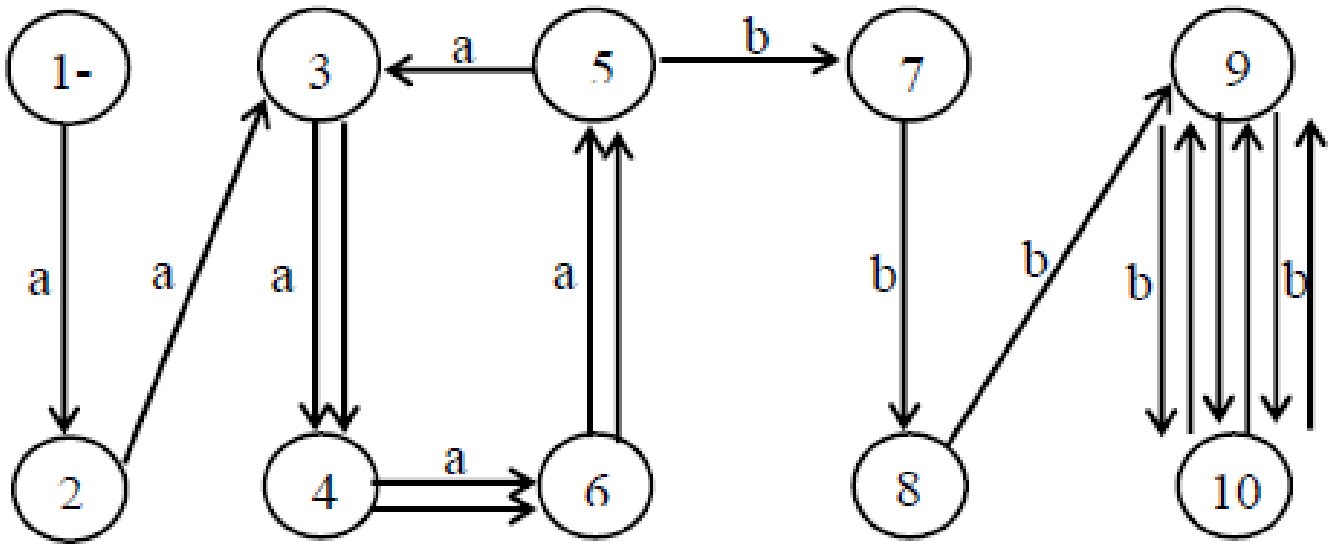
- The first 96 letter a's of this input string trace a path through this machine.
- The path cannot visit a new state when each input letter is read, because there are only 95 states. Therefore, at some point the path returns to a state that it has already visited.
- In other words, the path must contain a **circuit** in it. (A circuit is a loop that can be made of several edges).
- So, the path first wanders up to the circuit and then starts to loop around the circuit (maybe many times) until a b is read from the input.
- At this point, the path can take a different turn following the b-edges, and eventually end up at a final state where the word $a^{96}b^{96}$ is accepted.

Example 1:

- Just for the sake of argument again, let us say that the circuit that the a-edge path loops around has 7 states in it.
- Then what would happen to the input string $a^{96+7}b^{96}$?
- Just as in the case of the input string $a^{96}b^{96}$, the input string $a^{96+7}b^{96}$ would produce a path through the machine and loop around the circuit exactly in the same way, but **one more time** than the path produced by the input string $a^{96}b^{96}$.
- Both paths, at exactly the same state in the circuit, begin to branch off on the b-road.

Summary

- In summary, we can always choose a word in L that is large enough so that its path through the FA has to contain a circuit.
    - Once we find that some path with a circuit can reach a final state, we ask ourselves what happens to a path that is just like the first one, but that loops around the circuit one extra time and then proceeds identically through the machine.
    - The new path also leads to the same final state, but it is generated by a different input string which is **not** in the language L.
    - We then can conclude that there is no FA that can accepts all the words in L and only the words in L. Therefore, L is non-regular.
- This idea is called the **pumping lemma** discovered by Bar-Hillel, Perles, and Shamir in 1961. It is called "pumping" because we pump more letters into the middle of the words. It is called "lemma" because it is used as a tool to prove other results (i.e., certain specific languages are non-regular).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Theorem 13

- **Let L be any regular language that has infinitely many words. Then there exist some three strings x, y, and z (where y is not the null string) such that all the strings of the form**

$$xy^nz \text{ for } n = 1, 2, 3, \ldots$$

are words in L.

Proof

- Since L is regular, there is an FA that accepts exactly the words in L.
- Let *w* be some word in L that has more letters than there are states in FA.
- When *w* generates a path through the machine, the path cannot visit a new state for each letter read, because there are more letters than states. Therefore, the path must at some point revisit a state that it has already visited. In other words, the path contains a circuit in it.

Proof

Let's break the word *w* up into 3 parts:

1. Part 1: Starting at the beginning, let *x* denote all the letters of *w* that lead up to the **first** state that is revisited. Note that *x* may be the null string if the path revisits the start state as its first revisit.

2. Part 2: Starting at the letter after the substring *x*, let *y* denote the substring of w that travels around the circuit coming back to the same state the circuit began with. Because there must be a circuit, y cannot be the null string, and y contains the letters of w for exactly one loop around this circuit.

3. Part 3: Let *z* be the rest of *w*, starting at the letter after *y* and going to the end of the string *w*. Note that *z* could be null, or the path for *z* could also loop around the *y*-circuit or any other. That means that what *z* does is arbitrary.

- Clearly, from the definitions of these substrings, we have $w = xyz$. Recall that *w* is accepted by the FA.

Proof

- What is the path for the input string *xyyz*?
- This path follows the path for *w* in the first part *x* and leads up to the beginning of the place where w looped around a circuit.
- Then like *w*, it inputs the substring *y*, which causes the machine to loop back to this same state again.
- Then, again like *w*, it inputs the substring *y*, which causes the machine to loop back to this same state another time.
- Finally, just like *w*, it proceeds along the path dictated by the substring *z* and ends at the same final state that *w* did.
- Hence, the string *xyyz* is accepted by this machine and therefore must be in the language L.

Proof of Theorem

- Similarly, the strings xyyyz, xyyyyz, ... must also be in L.

- In other words, L must contain all strings of the form:

  xy^nz for n = 1; 2; 3; …

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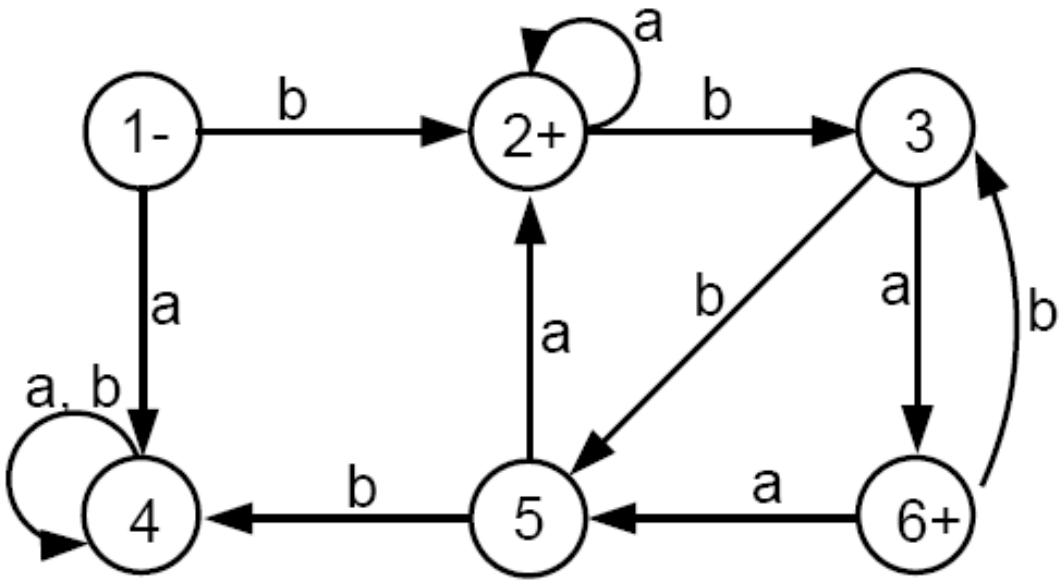
Proof of Theorem

- Consider the following FA that accepts an infinite language and has only six states:



- Consider the word $w = bbbababa$